

Adaptive User Representation Capacity in Deep Collaborative Filtering

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Abstract—Recommender systems aim to match users with items that best reflect their preferences. Collaborative filtering (CF) approaches typically learn fixed-dimensional embeddings for users and items and model their interactions to predict preferences. However, user preferences are often heterogeneous, and a fixed-capacity embedding may contain inactive or noisy latent dimensions for many users, which can impact interaction modeling negatively. In this paper, we investigate whether adaptive user representation capacity can improve recommendation performance. We implement baseline recommender models including Neural Matrix Factorization (NeuMF) and Deep Collaborative Filtering (DeepCF), and propose a masking mechanism that selectively activates relevant latent dimensions for each user before modeling user-item interactions. The performance of the proposed approach will be evaluated using ranking metrics such as Hit Rate (HR@K) and Normalized Discounted Cumulative Gain (NDCG@K) on benchmark recommendation datasets.

I. INTRODUCTION

Recommender systems play a crucial role in modern online platforms by providing personalized suggestions to users. Applications include movie recommendation, product recommendation in e-commerce, and music recommendation systems. Collaborative filtering techniques learn user preferences from historical interactions between users and items.

Traditional methods such as Matrix Factorization represent users and items in a shared latent space where recommendations are generated using similarity between embeddings. Recent deep learning approaches such as Neural Collaborative Filtering and Deep Collaborative Filtering (DeepCF) aim to learn more expressive interaction functions between users and items.

However, most recommender systems assume a fixed representation capacity for all users. In practice, users exhibit varying behavioral complexity. Some users interact with a narrow set of items, while others have diverse interests. Thus, representing all users with the same embedding capacity may negatively impact the performance of the recommender system. This project explores whether adaptive user representation capacity can improve recommendation performance by allowing the model to selectively activate relevant latent dimensions.

Although latent factor models have proven effective, they typically represent each user with a dense embedding vector of fixed dimensionality. This assumption may be limiting, as not all users require the same level of representational

complexity. For example, users with uniform or narrowly focused preferences may only need a small subset of latent factors, while users with diverse tastes may require richer representations. Enforcing the same embedding dimensionality for all users may therefore lead to suboptimal or weaker user representations.

Recent research has attempted to address these limitations by developing models that capture complex user behavior through deep architectures and multi-interest representations [1]. These methods highlight the importance of flexible representation learning in recommender systems. However, many existing approaches still rely on dense embeddings where all latent dimensions contribute equally during interaction modeling. In practice, only a subset of latent factors may be relevant for a given user when interacting with specific items.

Motivated by these observations, this work explores mechanisms that allow the model to dynamically control the contribution of each latent representation dimension during recommendation. Instead of assuming that every dimension of a user embedding is equally important, the model can learn to use only the most relevant latent factors while suppressing less useful ones. Such adaptive control over representation capacity may improve both the efficiency and expressiveness of collaborative filtering models.

To study this idea, we build upon deep collaborative filtering architectures and explore techniques that enable selective activation of user representation components during interaction modeling. By integrating adaptive representation mechanisms into existing recommender system frameworks, we aim to better align the learned representations with the underlying complexity of user preferences.

II. LITERATURE SURVEY

A. Matrix Factorization Based Collaborative Filtering

Matrix factorization has been one of the most influential approaches in collaborative filtering recommender systems. Koren et al. [2] gave the idea that user preferences and item characteristics can be represented in a shared latent factor space. In this framework, each user and item is associated with an embedding vector, and the predicted interaction between a user u and an item i is computed as the dot product of their latent vectors:

$$\hat{y}_{ui} = u^T v \quad (1)$$

This equation allows recommender systems to capture latent relationships between users and items that are not observable in raw interaction data. Matrix factorization methods gained popularity due to their scalability and strong empirical performance. However, these models rely on a linear interaction function and assume that all dimensions contribute equally to the prediction score. This assumption limits their ability to capture complex user-item relationships.

To address the limitations of pointwise prediction models, Rendle et al. [3] introduced Bayesian Personalized Ranking (BPR), a pairwise learning framework designed for implicit feedback datasets. Instead of predicting absolute interaction scores, BPR learns the relative ranking of items by assigning higher scores to positive items than to negative items. This objective aligns well with ranking-based evaluation metrics such as Hit Rate and Normalized Discounted Cumulative Gain.

B. Neural Collaborative Filtering

Recent advances in deep learning have motivated the development of neural collaborative filtering models that extend traditional latent factor approaches. He et al. [4] proposed Neural Collaborative Filtering (NCF), which replaces the linear dot-product interaction with a neural network-based interaction function. In NCF, user and item embeddings are concatenated and passed through a multilayer perceptron (MLP) that learns nonlinear feature interactions.

This neural architecture can definitely increase the expressiveness of the models and enable them to capture complex behavioral relationships that cannot be modeled using linear methods such as dot product. However, despite their increased modeling capacity, neural recommender systems still rely on dense embeddings with fixed dimensionality, which may contain redundant or inactive latent dimensions.

C. Deep Collaborative Filtering

Deep Collaborative Filtering (DeepCF) [5] was proposed to unify two complementary paradigms in recommender systems: representation learning and matching function learning. Representation learning approaches map users and items into a shared space where similarity reflects preference compatibility, while matching function learning approaches directly learn complex nonlinear mappings from user-item pairs to interaction scores.

DeepCF integrates these two methodologies by combining latent factor representations with deep neural matching functions. This unified framework improves recommendation accuracy by taking into consideration both representation similarity and nonlinear interaction modeling. However, similar to other deep recommendation models, DeepCF still utilizes dense embedding representations with fixed capacity, which may not adequately capture the changing complexity of user preferences.

D. Attention-Based Interaction Modeling

Attention mechanisms have led to improvement in interaction modeling in recommender systems. Attention-based models dynamically assign importance weights to different features. This enabling the model to focus on the most relevant aspects of user and item representations. AutoInt [6] utilizes self-attention to learn high-order feature interactions automatically. Thereby, demonstrating that attention mechanisms can effectively capture complex relationships between latent features.

Similarly, sequential recommendation models such as SAS-Rec [7] employ transformer-based architectures to model user behavior using self-attention mechanisms. While attention-based methods provide flexible interaction modeling capabilities, they typically operate on dense embeddings where all latent dimensions are treated equally. As a result, unnecessary or weakly activated dimensions may still influence attention scores, potentially introducing noise into the interaction modeling process.

E. Adaptive Representation and Sparse Embeddings

Recent work has explored adaptive representation learning techniques to better capture diverse user interests. The Multi-Interest Network with Dynamic Routing (MIND) model [1] proposes learning multiple embedding vectors for each user to represent different interest clusters which are derived from the user's interaction history. This approach tackles the limitation of representing each user with a single embedding vector.

In addition, sparse representation techniques have been known to improve the efficiency and interpretability of deep recommendation models. Sparse Neural Collaborative Filtering [8] introduces sparsity-inducing mechanisms that encourage the model to utilize only a subset of embedding dimensions, reducing redundancy in the latent space while maintaining recommendation accuracy.

These studies highlight the importance of adaptive and sparse representations in recommender systems. However, most existing approaches focus on either learning multiple representations or enforcing sparsity globally. Motivated by these observations, this work explores a masking-based model that selectively activates only the relevant user embedding dimensions before performing attention-based interaction modeling in deep collaborative filtering frameworks.

III. PRELIMINARIES

A. Problem Formulation

Consider a set of users $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ and a set of items $\mathcal{I} = \{i_1, i_2, \dots, i_N\}$. User preferences are represented through an interaction matrix $R \in \mathbb{R}^{M \times N}$, where each entry r_{ui} indicates whether user u has interacted with item i .

In many real-world recommender systems, explicit feedback such as ratings may be unavailable or sparse. Instead, recommendation models are often trained using implicit feedback derived from user behavior, such as clicks, purchases, or views. In this setting, interactions are typically represented as binary signals:

$$r_{ui} = \begin{cases} 1, & \text{if user } u \text{ has interacted with item } i \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

The objective of a recommender system is to learn a function $f(u, i)$ that predicts the likelihood of interaction between a user and an item:

$$\hat{y}_{ui} = f(u, i) \quad (3)$$

The learned scores $\hat{y}_{ui}$ are then used to rank candidate items for each user.

B. Collaborative Filtering with Latent Representations

Collaborative filtering methods model users and items using latent representations. Each user u is associated with an embedding vector $p_u \in \mathbb{R}^d$, and each item i is associated with an embedding vector $q_i \in \mathbb{R}^d$, where d denotes the embedding dimensionality.

Traditional matrix factorization models compute the interaction score using the dot product of the latent vectors:

$$\hat{y}_{ui} = p_u^T q_i \quad (4)$$

This formulation captures linear relationships between users and items and has been widely adopted in collaborative filtering systems.

More recent neural recommender models extend this formulation by learning nonlinear interaction functions between user and item embeddings. These models typically employ neural networks to capture complex interaction patterns that cannot be represented by simple dot products.

C. Limitations of Fixed Representation Capacity

Most collaborative filtering models represent users with embedding vectors of fixed dimensionality. This implies that all users share the same representational capacity regardless of their behavioral complexity.

However, user interaction patterns can vary significantly. Some users exhibit focused preferences and interact with a small number of item categories, while others display diverse interests across multiple domains. In such cases, a fixed-dimensional dense representation may either underrepresent complex users or allocate unnecessary capacity to simpler users.

This observation suggests that different users may benefit from different levels of representational complexity. Ideally, the recommendation model should be able to emphasize the most relevant latent factors for each user while suppressing less informative dimensions.

D. Problem Statement

Given a set of users $\mathcal{U}$, items $\mathcal{I}$, and an interaction matrix R , the goal of this work is to design a collaborative filtering model that can adaptively control the effective representation capacity of user embeddings during interaction modeling.

Specifically, we investigate mechanisms that allow the model to dynamically adjust the contribution of latent representation dimensions when predicting user-item interactions. By enabling selective activation of latent factors, the model can better align the learned user representations with the underlying structure of user preferences.

The effectiveness of the proposed approach will be evaluated by comparing recommendation performance against established collaborative filtering baselines using standard ranking metrics.

IV. BASELINE MODELS

To evaluate the performance of the proposed adaptive representation mechanism, we compare our model against two widely used neural collaborative filtering architectures. These models represent standard approaches for learning user-item interactions using neural networks.

A. Neural Collaborative Filtering (NCF)

Neural Collaborative Filtering (NCF) [4] replaces the traditional dot-product interaction used in matrix factorization with a neural interaction function. In this framework, each user u and item i is represented by a learnable embedding vector p_u and q_i in a shared latent space.

The embeddings are concatenated and passed through a multilayer perceptron (MLP) to model nonlinear user-item interactions:

$$\hat{y}_{ui} = f_{\text{MLP}}([p_u, q_i]) \quad (5)$$

B. Deep Collaborative Filtering (DeepCF)

Deep Collaborative Filtering (DeepCF) [5] extends neural collaborative filtering by combining two complementary interaction modeling strategies: representation learning and matching function learning.

The representation learning component models similarity between user and item embeddings through element-wise interactions:

$$z_{RL} = p_u \odot q_i \quad (6)$$

The matching function learning component models nonlinear relationships using a multilayer perceptron:

$$z_{ML} = f_{\text{MLP}}([p_u, q_i]) \quad (7)$$

The outputs of these two components are fused to produce the final interaction score. This architecture captures both similarity and complex interaction patterns between users and items.

DeepCF serves as the primary baseline in our experiments, as it represents a strong neural collaborative filtering model upon which our proposed adaptive representation mechanism builds.

The performance of the proposed model is compared against these baselines using ranking metrics including Hit Rate (HR@10) and Normalized Discounted Cumulative Gain (NDCG@10).

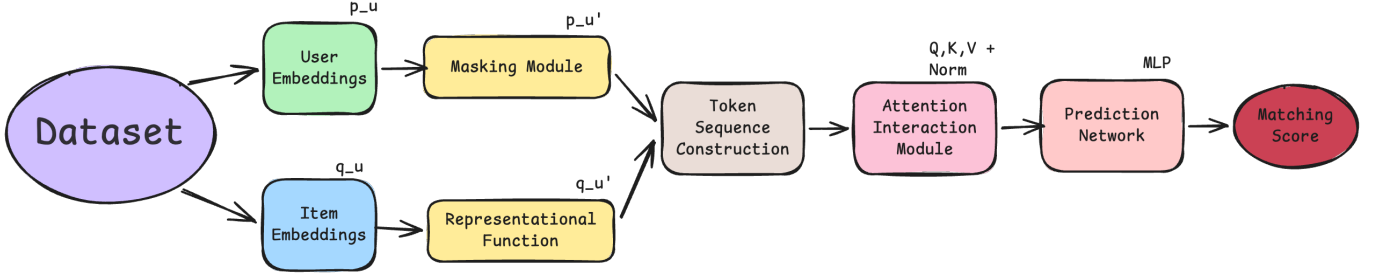


Fig. 1. Workflow of the Proposed Masked Attention CF

V. PROPOSED METHODOLOGY

Standard neural recommender systems typically feed raw user and item embeddings directly into an interaction module (e.g., an MLP or inner product). In sparse datasets, however, this naive approach suffers from *information dilution*: many embedding dimensions remain near their random initialization and inject noise into the interaction signal. To address this, we propose **RP-UCB** (Relative Parametric Upper Confidence Bound), a novel mechanism that adaptively controls the effective representation capacity of each user by selectively activating only the most reliable latent dimensions.

A. Problem Setup and Notation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, with $|\mathcal{U}| = M$, $|\mathcal{I}| = N$. The interaction matrix $R \in \mathbb{R}^{M \times N}$ contains binary implicit feedback constructed via thresholding:

$$y_{ui} = \mathbb{I}(r_{ui} \geq \tau), \quad (8)$$

where r_{ui} is the original rating (if available) and τ a predefined threshold. For each user u , let $N_u = \sum_i y_{ui}$ be the number of positive interactions. Denote by $\bar{N} = \frac{1}{M} \sum_u N_u$ the average number of interactions per user.

Each user u and item i is associated with learnable embeddings:

$$p_u = \text{Embed}(u, E_P) \in \mathbb{R}^d, \quad (9)$$

$$q_i = \text{Embed}(i, E_I) \in \mathbb{R}^d, \quad (10)$$

where E_P and E_I are embedding matrices. Additionally, we introduce a user-specific mask weight vector $w_u \in \mathbb{R}^d$ obtained from a learnable matrix E_W :

$$w_u = \text{Embed}(u, E_W) \in \mathbb{R}^d. \quad (11)$$

B. Motivation: Gradient Collapse for Sparse Users

We first analyze the behavior of a static mask $m_u = \sigma(w_u)$ used to gate the user embedding:

$$p_u^{(\text{masked})} = p_u \odot m_u, \quad (12)$$

where $\sigma(\cdot)$ is the sigmoid function applied element-wise. The interaction score $\hat{y}_{ui}$ is computed by a differentiable function f (e.g., an MLP) on the masked user embedding and the item embedding: $\hat{y}_{ui} = f(p_u \odot m_u, q_i)$.

Training employs the Bayesian Personalized Ranking (BPR) loss [3]:

$$\mathcal{L}_{\text{BPR}} = - \sum_u \sum_{i \in \mathcal{I}_u^+} \sum_{j \in \mathcal{I}_u^-} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \quad (13)$$

where $\mathcal{I}_u^+$ are positive items, $\mathcal{I}_u^-$ sampled negatives. For a single positive-negative pair (i, j) , the gradient with respect to $w_{u,k}$ (the k -th component of w_u) is:

$$\frac{\partial \mathcal{L}_{\text{BPR}}}{\partial w_{u,k}} = -(1 - \sigma(\hat{y}_{ui} - \hat{y}_{uj})) p_{u,k} \sigma'(w_{u,k}) (q_{i,k} - q_{j,k}). \quad (14)$$

When N_u is very small, the negative samples j exhibit high variance, making the term $(q_{i,k} - q_{j,k})$ noisy. To minimize the loss, the optimizer drives $w_{u,k} \rightarrow -\infty$, causing $m_{u,k} = \sigma(w_{u,k}) \rightarrow 0$. This phenomenon, termed *representation collapse*, renders the user embedding effectively zero for many dimensions, losing valuable information.

C. Static Masking: Insufficient for Sparse Users

A direct application of the learned mask $m_u = \sigma(w_u)$ (without any regularization) fails for cold-start users because the gradient noise forces the mask to zero. To counter this, we first introduce a UCB-inspired exploration term that keeps the mask open for under-explored users.

D. Dynamic Uncertainty-Aware Masking (UCB)

Inspired by the Upper Confidence Bound (UCB) principle from bandit theory, we augment the static mask with an exploration bonus:

$$m_u^{(\text{UCB})} = \min \left(1, \sigma(w_u) + \frac{\beta}{\sqrt{N_u} + \epsilon} \right), \quad (15)$$

where $\beta > 0$ is an exploration hyperparameter and ϵ a small constant to avoid division by zero. For sparse users (N_u small), the bonus $\beta/\sqrt{N_u}$ is large, forcing the mask to be positive; as N_u grows, the bonus decays, relying more on the learned $\sigma(w_u)$. However, this formulation still has two drawbacks:

- 1) The bonus applies equally to all dimensions, even those known to be unreliable.
- 2) The decay $1/\sqrt{N_u}$ never reaches exactly zero, so the mask never fully relies on the learned weights, even for users with abundant data.

E. RP-UCB: Relative Parametric Upper Confidence Bound

To overcome these limitations, we propose **RP-UCB**, a unified mechanism that introduces a *dataset-relative* exploration term and a *dimension-specific confidence* gate.

We introduce a learnable confidence vector $\gamma \in \mathbb{R}^d$ (shared across users) and define the final adaptive mask as:

$$m_u^{(\text{RP-UCB})} = \text{clamp}\left(\sigma(w_u) + \sigma(\gamma) \odot \max(0, \ln(\bar{N}/N_u)), 0, 1\right) \quad (16)$$

Here $\odot$ denotes element-wise multiplication, $\sigma(\gamma)$ is the sigmoid applied element-wise to γ , and $\text{clamp}(\cdot, 0, 1)$ projects each component into $[0, 1]$.

The term $\max(0, \ln(\bar{N}/N_u))$ acts as a *when to explore* scalar:

- If $N_u < \bar{N}$, then $\ln(\bar{N}/N_u) > 0$ and exploration is active.
- If $N_u \geq \bar{N}$, the $\max(0, \cdot)$ yields zero, completely disabling exploration.

The vector $\sigma(\gamma)$ learns *where to explore*: dimensions with $\sigma(\gamma_k) \approx 1$ receive the full exploration boost, while those with $\sigma(\gamma_k) \approx 0$ are never forced open, preventing noise from unreliable factors.

The resulting masked user embedding is $p_u^{(\text{masked})} = p_u \odot m_u^{(\text{RP-UCB})}$, which is then fed into any differentiable interaction function f (e.g., MLP, inner product, or even an attention module) to produce the final score $\hat{y}_{ui} = f(p_u^{(\text{masked})}, q_i)$.

F. Theoretical Properties of RP-UCB

We now state four key properties of RP-UCB that guarantee its effectiveness. Proofs are deferred to Section VI.

Lemma 1 (Boundedness): For every user u and dimension k , $0 \leq m_{u,k}^{(\text{RP-UCB})} \leq 1$.

Lemma 2 (Phase Transition): If $N_u \geq \bar{N}$, then $m_u^{(\text{RP-UCB})} = \sigma(w_u)$; i.e., the exploration term vanishes and the mask reduces to the static learned mask.

Lemma 3 (Collapse Prevention for Sparse Users): For any user u with $N_u < \bar{N}$ and any dimension k for which $\sigma(\gamma_k) > 0$,

$$m_{u,k}^{(\text{RP-UCB})} \geq \sigma(\gamma_k) \ln\left(\frac{\bar{N}}{N_u}\right) > 0. \quad (17)$$

Hence the mask is strictly positive on all reliable dimensions, preventing gradient-driven collapse.

Lemma 4 (Monotonic Decay of Exploration): For a fixed γ , the exploration bonus $\Delta_k(N_u) = \sigma(\gamma_k) \max(0, \ln(\bar{N}/N_u))$ is a non-increasing function of N_u , and $\Delta_k(N_u) \rightarrow 0$ as $N_u \rightarrow \bar{N}^-$.

G. Training Objective

We train the model end-to-end using the BPR loss combined with an ℓ_1 regularization on the mask values to encourage sparse activation:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda \sum_u \|m_u^{(\text{RP-UCB})}\|_1, \quad (18)$$

where λ controls the sparsity. The final mask is clamped to $[0, 1]$, and the ℓ_1 penalty promotes using only the most relevant dimensions.

H. Architectural Flexibility

RP-UCB is architecture-agnostic: the interaction function f can be any differentiable model. In our experiments, we evaluate RP-UCB on two popular collaborative filtering backbones:

- **DeepCF** [5]: combines element-wise product and MLP branches, fused for the final score.
- **DeepCF with Self-Attention**: replaces the fusion branch with a self-attention layer over the concatenated masked user and item embeddings, followed by an MLP.

This demonstrates that RP-UCB consistently improves performance regardless of the underlying interaction model.

Algorithm 1 Training RP-UCB

Require: Training interactions $\mathcal{D}$, embedding dimension d , hyperparameters β, λ, τ

Ensure: Trained embeddings E_P, E_W, E_I and confidence vector γ

- 1: Compute average interactions $\bar{N} = \frac{1}{|\mathcal{I}|} \sum_u N_u$
 - 2: Initialize E_P, E_W, E_I, γ randomly
 - 3: **for** each epoch **do**
 - 4: **for** each batch $\mathcal{B} \subseteq \mathcal{D}$ **do**
 - 5: **for** each $(u, i, j) \in \mathcal{B}$ **do**
 - 6: Retrieve $p_u \leftarrow E_P[u]$, $w_u \leftarrow E_W[u]$, $q_i \leftarrow E_I[i]$, $q_j \leftarrow E_I[j]$
 - 7: Compute static mask $m_u^{\text{static}} = \sigma(w_u)$
 - 8: Compute exploration scalar $e = \max(0, \ln(\bar{N}/N_u))$
 - 9: Compute RP-UCB mask $m_u = \text{clamp}(m_u^{\text{static}} + \sigma(\gamma) \odot e, 0, 1)$
 - 10: Mask user embeddings: $\tilde{p}_u = p_u \odot m_u$
 - 11: Compute scores: $\hat{y}_{ui} = f(\tilde{p}_u, q_i)$, $\hat{y}_{uj} = f(\tilde{p}_u, q_j)$
 - 12: Compute BPR loss $\ell_{\text{BPR}} = -\log \sigma(\hat{y}_{ui} - \hat{y}_{uj})$
 - 13: Compute regularization $\ell_{\text{reg}} = \lambda \|m_u\|_1$
 - 14: Accumulate total loss $\ell = \ell_{\text{BPR}} + \ell_{\text{reg}}$
 - 15: **end for**
 - 16: Perform backward pass and update E_P, E_W, E_I, γ
 - 17: **end for**
 - 18: **end for**
-

VI. PROOFS OF THEORETICAL PROPERTIES

We now provide rigorous proofs for the lemmas introduced in Section V. The following notation is used throughout: $m_{u,k}^{(\text{RP-UCB})}$ denotes the k -th component of the mask for user u , $\sigma(\cdot)$ is the sigmoid function, $\text{clamp}(x, 0, 1) = \min(\max(x, 0), 1)$, and $\odot$ represents element-wise multiplication.

A. Proof of Lemma 1 (Boundedness)

[Boundedness] For every user u and every dimension k ,

$$0 \leq m_{u,k}^{(\text{RP-UCB})} \leq 1.$$

Proof 1: By definition,

$$m_u^{(\text{RP-UCB})} = \text{clamp}(\sigma(w_u) + \sigma(\gamma) \odot \max(0, \ln(\bar{N}/N_u)), 0, 1).$$

The clamp function explicitly bounds each component to the interval $[0, 1]$. Hence the claim holds.

B. Proof of Lemma 2 (Phase Transition)

[Phase Transition] If $N_u \geq \bar{N}$, then

$$m_u^{(\text{RP-UCB})} = \sigma(w_u).$$

Proof 2: When $N_u \geq \bar{N}$, we have $\bar{N}/N_u \leq 1$, so $\ln(\bar{N}/N_u) \leq 0$. The $\max(0, \cdot)$ function then returns 0. Therefore,

$$\sigma(w_u) + \sigma(\gamma) \odot 0 = \sigma(w_u).$$

Applying the clamp, $\text{clamp}(\sigma(w_u), 0, 1)$. Since $\sigma(w_u) \in (0, 1)^d$ component-wise, the clamp leaves it unchanged. Thus $m_u^{(\text{RP-UCB})} = \sigma(w_u)$.

C. Proof of Lemma 3 (Collapse Prevention for Sparse Users)

[Collapse Prevention for Sparse Users] For any user u with $N_u < \bar{N}$ and any dimension k for which $\sigma(\gamma_k) > 0$, we have

$$m_{u,k}^{(\text{RP-UCB})} \geq \sigma(\gamma_k) \ln\left(\frac{\bar{N}}{N_u}\right) > 0.$$

Consequently, the mask is strictly positive on all reliable dimensions, preventing gradient-driven collapse.

Proof 3: Since $N_u < \bar{N}$, we have $\ln(\bar{N}/N_u) > 0$. Define $\Delta_k = \sigma(\gamma_k) \ln(\bar{N}/N_u) > 0$. Then the argument of the clamp is

$$\alpha_k = \sigma(w_{u,k}) + \Delta_k.$$

Because $\sigma(w_{u,k}) \geq 0$, we have $\alpha_k \geq \Delta_k > 0$. The clamp function satisfies $\text{clamp}(x, 0, 1) \geq \min(x, 1)$. For $x \geq \Delta_k > 0$, we have $\text{clamp}(x, 0, 1) \geq \Delta_k$ (if $x \geq \Delta_k$, the clamp cannot reduce the value below Δ_k because it only caps at 0 from below and 1 from above; since $\Delta_k \leq 1$ for typical values, the lower bound holds). Therefore,

$$m_{u,k}^{(\text{RP-UCB})} \geq \Delta_k = \sigma(\gamma_k) \ln\left(\frac{\bar{N}}{N_u}\right) > 0.$$

The positivity ensures that the effective user embedding dimension remains active, avoiding the collapse observed in the static mask.

D. Proof of Lemma 4 (Monotonic Decay of Exploration)

[Monotonic Decay of Exploration] For a fixed confidence vector γ , define the exploration bonus on dimension k as

$$\Delta_k(N_u) = \sigma(\gamma_k) \max\left(0, \ln\left(\frac{\bar{N}}{N_u}\right)\right).$$

Then $\Delta_k(N_u)$ is a non-increasing function of N_u for $N_u > 0$, and $\Delta_k(N_u) \rightarrow 0$ as $N_u \rightarrow \bar{N}^-$.

Proof 4: For $N_u \geq \bar{N}$, $\Delta_k(N_u) = 0$ and is constant. For $0 < N_u < \bar{N}$, we have $\Delta_k(N_u) = \sigma(\gamma_k) \ln(\bar{N}/N_u)$. Differentiating with respect to N_u ,

$$\frac{d}{dN_u} \Delta_k(N_u) = \sigma(\gamma_k) \cdot \frac{d}{dN_u} (\ln \bar{N} - \ln N_u) = -\frac{\sigma(\gamma_k)}{N_u} < 0.$$

Thus $\Delta_k(N_u)$ is strictly decreasing on $(0, \bar{N})$. Since $\Delta_k(N_u)$ is continuous from the left at $\bar{N}$ and $\lim_{N_u \rightarrow \bar{N}^-} \ln(\bar{N}/N_u) = 0$, we have $\Delta_k(N_u) \rightarrow 0$ as $N_u \rightarrow \bar{N}^-$. Hence the bonus decays monotonically to zero as the user approaches the average interaction count.

REFERENCES

- [1] C. Li, Z. Liu, M. Wu, Y. Xu, H. Zhao, P. Huang, G. Kang, Q. Chen, W. Li, and D. Lee, "Mind: A multi-interest network with dynamic routing for recommendation at tmall," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2019, pp. 2617–2625.
- [2] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *IEEE Computer*, vol. 42, no. 8, pp. 30–37, 2009.
- [3] S. Rendle, C. Freudenthaler, Z. Gantner, and L. Schmidt-Thieme, "Bpr: Bayesian personalized ranking from implicit feedback," in *Proceedings of the 25th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2009, pp. 452–461.
- [4] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, "Neural collaborative filtering," in *Proceedings of the 26th International World Wide Web Conference (WWW)*, 2017, pp. 173–182.
- [5] Z. Deng, Y. Yan, Y. Li, and J. Tang, "Deep collaborative filtering: A unified framework of representation learning and matching function learning in recommender systems," in *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, 2018, pp. 2780–2786.
- [6] W. Song, C. Shi, Z. Xiao, Z. Duan, Y. Xu, M. Zhang, and J. Tang, "Autoint: Automatic feature interaction learning via self-attentive neural networks," in *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, 2019, pp. 1161–1170.
- [7] W.-C. Kang and J. McAuley, "Self-attentive sequential recommendation," in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2018, pp. 197–206.
- [8] S. Zhang, L. Yao, A. Sun, and Y. Tay, "Sparse neural collaborative filtering," *arXiv preprint arXiv:1806.01747*, 2018.